Chapter 17, Problem 58P

Problem

Find the exponential Fourier series of a function that has the following trigonometric Fourier series coefficients:

$$a_0 = \frac{\pi}{4}, \qquad b_n = \frac{(-1)^n}{n}, \qquad a_n = \frac{(-1)^n - 1}{\pi n^2}$$

Take $T = 2\pi$.

Step-by-step solution

Step 1 of 1

$$T = 2\pi \qquad (given)$$

$$C_o = a_o = \frac{\pi}{4}$$

$$C_n = \frac{a_n - jb_n}{2}$$

$$= \frac{(-1)^n - 1}{n^2\pi} - \frac{j(-1)^n}{n}$$

$$f(t) = C_o + \sum_{n=-\infty}^{\infty} C_n\, e^{-jn\omega_o t}$$

$$\omega_o = \frac{2\pi}{T} = 1$$

$$\boxed{f(t) = \frac{\pi}{4} + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \left[\frac{(-1)^n - 1}{n^2\pi} - \frac{j(-1)^n}{n}\right] e^{-jnt}}$$